

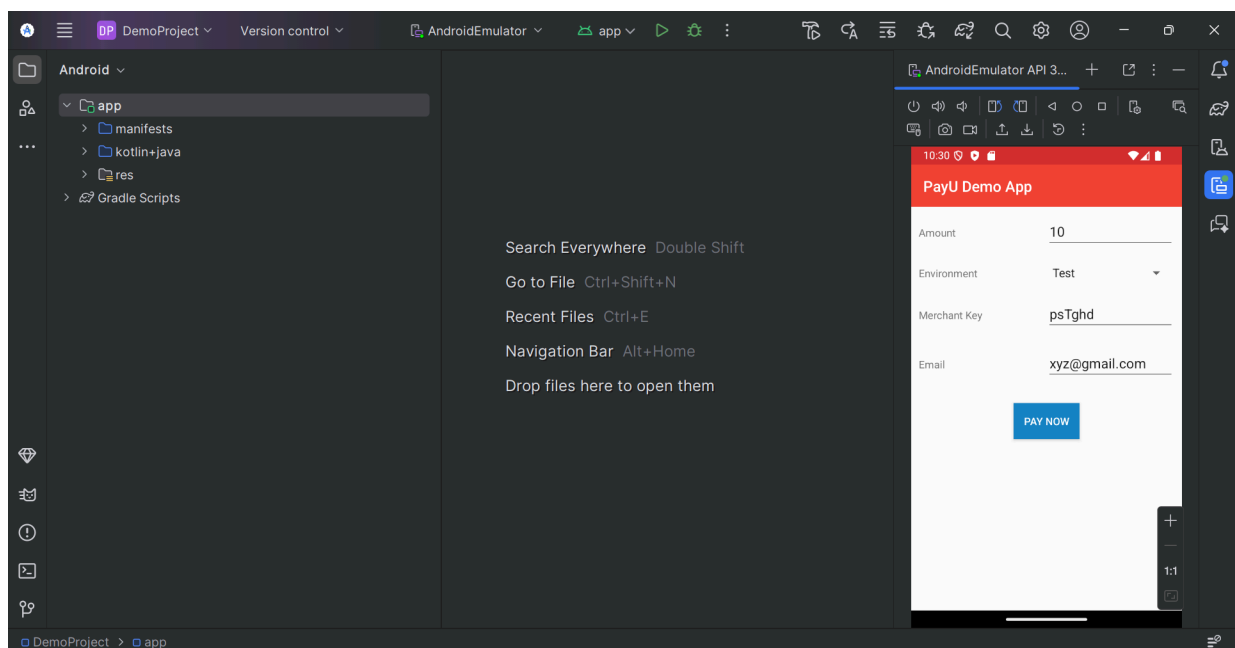
1. Android Emulator (Virtual Device) Setup

Creating Virtual Device

- Opened **Android Studio** → AVD Manager.
- Created a new Virtual Device with the required Android version.

Verification

- Launched Emulator successfully.
- Verified device connection using:



Purpose: Emulator is used for testing without a physical device.

2. Real Android Device Configuration

Device Preparation

- Enabled **Developer Options** on Android device.
- Enabled **USB Debugging**.

Connection

- Connected device to system via USB cable.

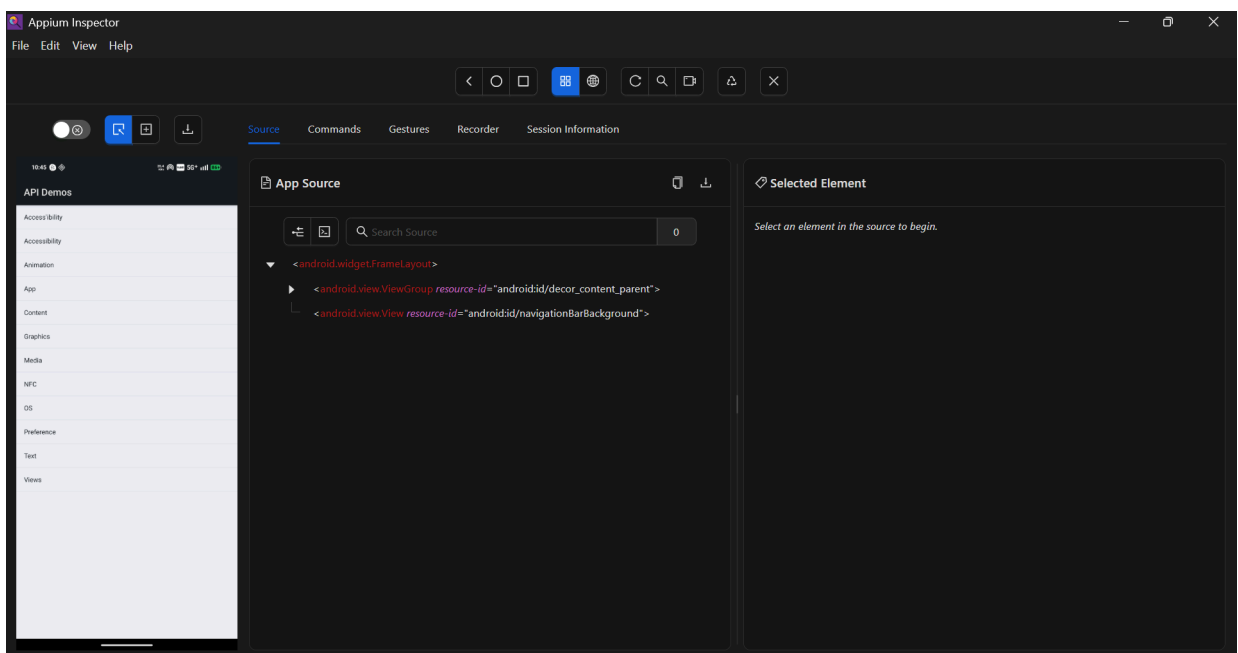
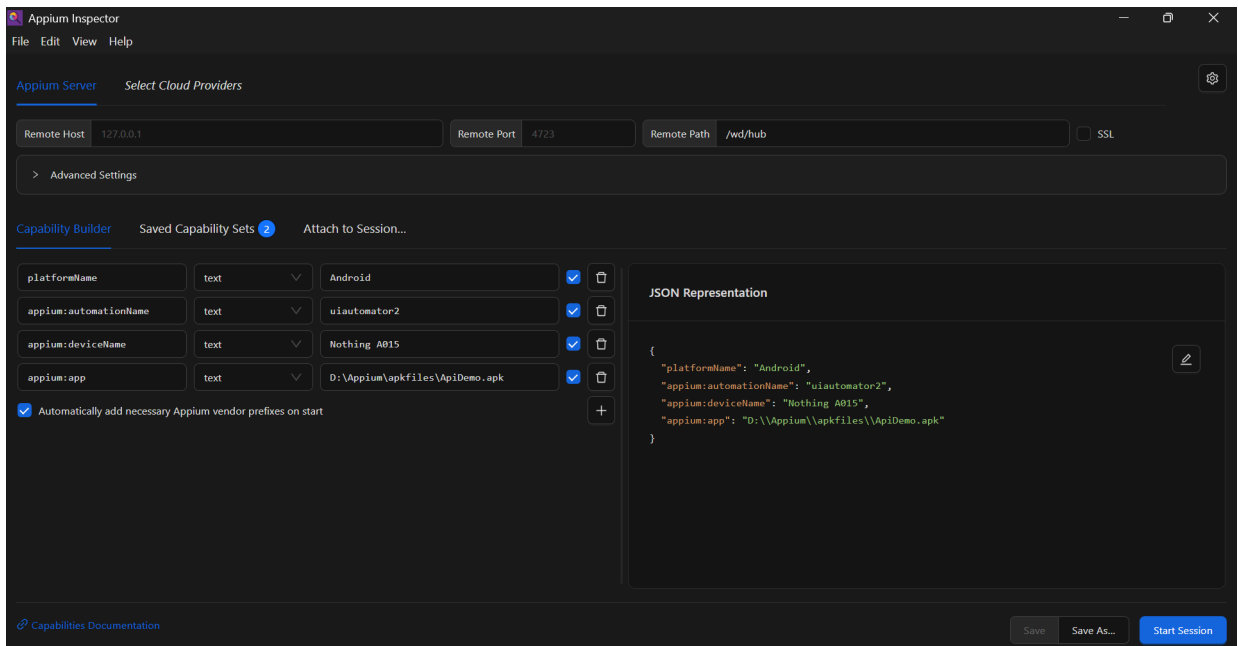
Verification

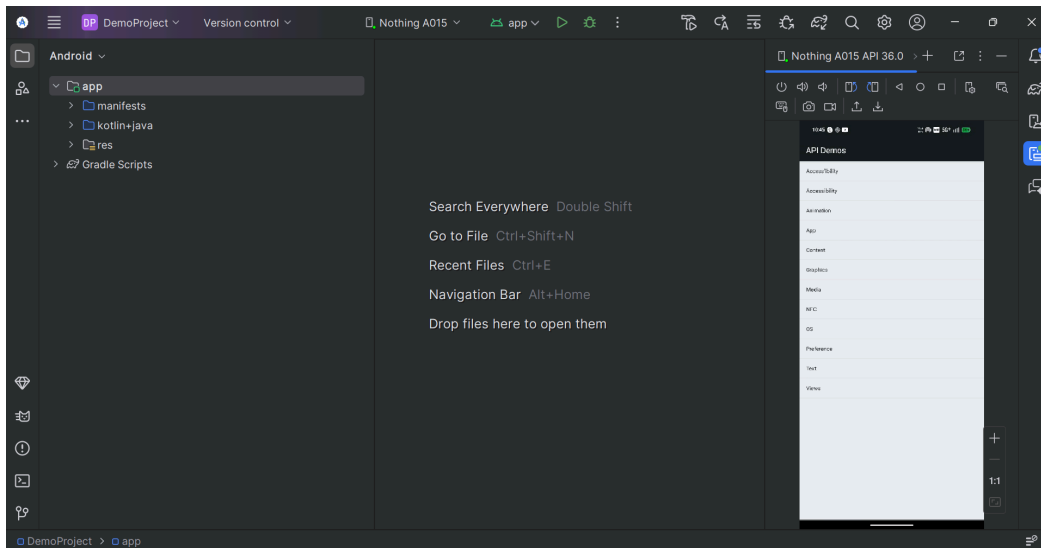
- Verified connection using:

adb devices

Authorization

- Allowed USB debugging permission on the device when prompted.





Purpose: Real device testing ensures real-world behavior validation.

3. Installing APK on Real Device

- The same APK file was installed on the real Android device.
- Installation was done using:
ADB commands, or
Automation code written inside the Maven project.
- The application was successfully installed and launched on the real device.

ADB Command Used

```
C:\Windows\System32\cmd.e x + v
Microsoft Windows [Version 10.0.26200.7705]
(c) Microsoft Corporation. All rights reserved.

D:\Appium\apkfiles>adb.exe install ApiDemo.apk
Performing Streamed Install
Success

D:\Appium\apkfiles>
```

Purpose: Ensures the application is ready for automation testing on a real device.